Workshop – Week 5 – Some selected solutions

1. Write down any changes to registers, the memory locations and the program counter when the following MASSEY machine programs are run:

Program (a):

```
130F
8302
4383
E000
```

Solution to Q1(a):

Memory location 0 = 130F
Memory location 1 = 8302
Memory location 2 = 4383
Memory location 3 = E000

Program counter starts at 0.
130F – load R3 with 000F. Program Counter changes to 1.
8302 – shift R3 right by 2 bits. R3 is now 0003. Program Counter changes to 2.
4383 – store R3 in memory location 83. Memory location 83 is now 0003. PC changes to 3.
E000 – halt.

Program (b):

```
11E0
9108
4103
D1DD
```

Solution to Q1(b):

Memory location 0 = 11E0
Memory location 1 = 9108
Memory location 2 = 4103
Memory location 3 = D1DD

Program counter starts at 0.
11E0 – load R1 with 00E0. Program Counter changes to 1.
9108 – shift R1 left by 8 bits. R1 is now E000. Program Counter changes to 2.
4103 – store R1 in memory location 3. Memory location 3 is now E000. PC changes to 3.
E000 – halt.

Interesting notes:

a) a memory location can hold 16 bits (4 hex digits). So when you shift to the left by 8 bits, the effect is to shift the two right-most digits to the left-hand-side of the number. Eg: 00E0 becomes E000.

b) the instruction D1DD disappears because the new instruction E000 is loaded into the same memory location (memory location 3).

2. Write the following program as assembler instructions. This will make it easier to read.

```
1032
11FF
2071
9108
B117
C201
1301
6423
6504
E000
```

Solution to Q2:

LOAD R0 with 32
LOAD R1 with 00FF
LOAD R7 with R1 (R7 is now 00FF)
SHIFT R1 left 8 bits (R1 is now FF00)
R1 = R1 OR R7 (R1 is now FFFF)
R2 = R0 XOR R1 (R2 is now complement of 32)
LOAD R3 with 01
R4 = R2 + R3 (R4 is now complement of 32 plus 1, i.e. R4 = -32)
R5 = R0 + R4 (R5 = 32 + -32. R5 should be zero.)
HALT

3. Use the assembler code in (2) above to identify what is happening in the program.

a) What value is stored in register 5 and why?

b) Can you write the program to achieve the same result in a different way and thereby use less machine code instructions?

Solution to Q3(a):

R5 will have zero stored in it. The program simulates the twos complement process by creating the complement of 32 (and adding 1 to it) and then adding the result to 32.

Solution to Q3(b):

```
1032
2020
7020
6502
E000
```

4. Write a MASSEY machine code program that is the equivalent of the following C++ code:

```
i = 9;
while (i >= 0) {
  i--;
}
```

Solution to Q4:

There are many possible solutions. Here is one of the solutions:

```
1080
9008
1209
4247
1A01
70A0
3347
A230
D20C
633A
4347
D006
E000
```

Some notes on the way to producing the final machine code program:

LOAD R0 with 80
SHIFT R0 left 8 bits (R0 now has a mask of 1000 0000)

LOAD R2 with 9 (the start value for variable `i`)
STORE R2 in location 47 (use location 47 as variable `i`)

LOAD RA with 1
NEGATE RA (RA now has -1 stored in it)

[the test]
LOAD R3 with location 47
R2 = R3 AND R0 (will have 1 in the sign bit if negative)
JUMP if R2 == R0 to outside loop

R3 = R3 + RA
STORE R3 in location 47
JUMP if R0 == R0 to start of loop test

[outside the loop]
HALT